

Adaptive Sector Allocation via Regime-Switching: A Walk-Forward Backtest on SPDR Sector ETFs (2007–2024)

Cong-Hieu LE, Marilou BARBET
Master in Finance and Banking · UPF-BSM

Executive Summary. This report evaluates three sector portfolio strategies on the eleven SPDR sector ETFs over 2007-2024: minimum-variance (MV), maximum-Sharpe (MS), and a regime-switching portfolio (RS) that selects between the two based on a walk-forward KMeans classifier trained on realised volatility, momentum, and VIX. All strategies are estimated quarterly on a two-year trailing window with no look-ahead, and net of 10 bps transaction costs. The RS portfolio achieves the best Sharpe ratio (0.437) and Calmar ratio (0.209) among all strategies, outperforming both SPY and the two static approaches. A look-ahead comparison further shows that standard in-sample Markowitz substantially overstates achievable Sharpe ratios.

1. Introduction

Mean-variance optimisation [1] is the canonical framework for portfolio construction, but its out-of-sample performance is well-documented to be poor: estimation error in both means and covariances produces unstable weights [2]. A second issue is stationarity: Markowitz assumes fixed return distributions, whereas financial markets alternate between regimes with different volatility, correlation, and momentum properties.

This report addresses both by embedding Markowitz within a regime-detection layer. The central question is:

Does a regime-switching sector portfolio - selecting Max-Sharpe in bull markets and Min-Variance otherwise, based on a walk-forward KMeans classifier - generate superior risk-adjusted returns compared to static strategies and the S&P 500 over 2007-2024?

2. Data and Methodology

2.1. Universe and data

The investment universe consists of the eleven Select Sector SPDR ETFs (XLK, XLF, XLV, XLE, XLI, XLB, XLY, XLP, XLU, XLRE, XLC), covering all GICS sectors. Daily adjusted prices are sourced from Yahoo Finance (January 2005 - December 2024). SPY is the benchmark. XLRE and XLC enter the universe dynamically once they accumulate 504 trading days of history, avoiding look-forward bias.

2.2. Walk-forward backtest

Portfolio weights are rebalanced quarterly. At each rebalancing date t , only the trailing 504 trading days (≈ 2 years) are used for estimation. Weights are held fixed for the following quarter.

Transaction costs of 10bps are applied to turnover on each rebalancing day.

2.3. Portfolio strategies

Min-Variance (MV) minimises portfolio variance:

$$\min_{\mathbf{w}} \mathbf{w}^\top \hat{\Sigma} \mathbf{w} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad 0 \leq w_i \leq 0.30 \quad \forall i \quad (1)$$

where $\hat{\Sigma}$ is the $n \times n$ sample covariance matrix estimated on the trailing window.

Max-Sharpe (MS) maximises the Sharpe ratio:

$$\max_{\mathbf{w}} \frac{\mathbf{w}^\top \hat{\boldsymbol{\mu}} - r_f}{\sqrt{\mathbf{w}^\top \hat{\Sigma} \mathbf{w}}} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad 0 \leq w_i \leq 0.30 \quad \forall i \quad (2)$$

where $\hat{\boldsymbol{\mu}} \in R^n$ is the vector of sample mean daily returns and $r_f = 2\%$ annualised. Both are solved via SLSQP (Sequential Least Squares Programming), a numerical optimisation algorithm that handles nonlinear objectives with equality and inequality constraints.

The **Regime-Switching (RS)** portfolio selects between the two:

$$\mathbf{w}_t^{\text{RS}} = \begin{cases} \mathbf{w}_t^{\text{MS}} & \text{if regime}_t = \text{bull} \\ \mathbf{w}_t^{\text{MV}} & \text{otherwise} \end{cases} \quad (3)$$

2.4. Regime detection

At each rebalancing date, KMeans ($K = 3$) is fitted on the trailing 504-day window using three 21-day rolling features of SPY: annualised realised volatility, cumulative momentum, and VIX level. Features are standardised. Clusters are labelled automatically by mean momentum: **bull** (highest), **neutral**, **bear** (lowest). The model is re-estimated from scratch at every rebalancing date-no future data ever enters the classification.

3. Results

3.1. Cumulative performance

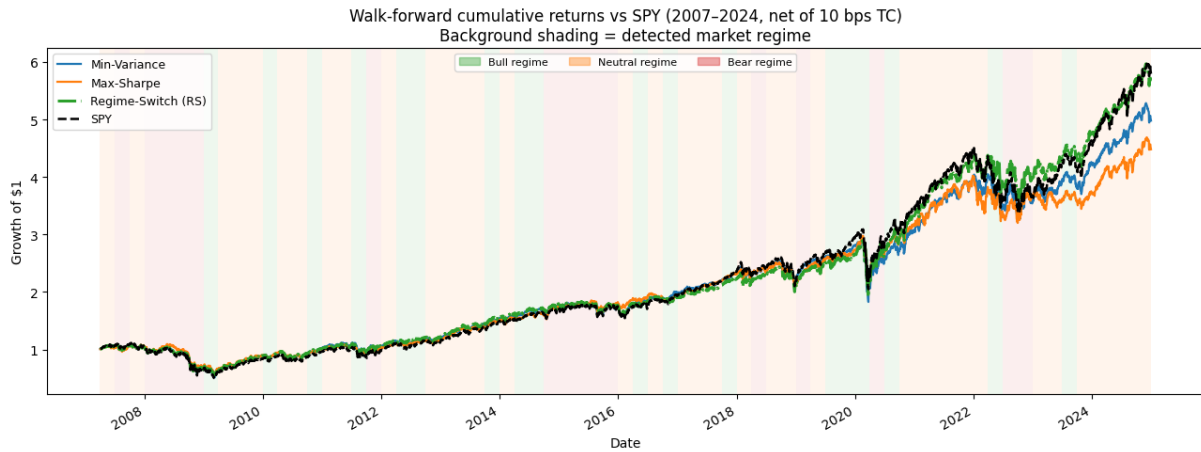


Figure 1: Walk-forward cumulative returns of Min-Variance, Max-Sharpe, Regime-Switch, and SPY (2007-2024, net of 10 bps TC). Background shading: green = bull regime, orange = neutral, red = bear.

Figure 1 reveals several key patterns. First, the three **crisis episodes** - 2008-2009, March 2020, and 2022 - are correctly identified as bear regimes (red shading), confirming that the KMeans classifier captures severe dislocations in real time. Second, the **2013-2019 expansion** is predominantly labelled bull, during which Max-Sharpe and Regime-Switch both track the upside of SPY, while Min-Variance visibly lags. Third, the **2022 transition** illustrates the inherent lag of a backward-looking classifier: the detector reacts, it does not predict. Finally, the **post - 2020 bull market** rewards the risk-on tilt of RS and MS, consistent with concentration in high-momentum sectors (XLK, XLC).

3.2. Performance metrics

Metric	Min-Variance	Max-Sharpe	Regime-Switch	SPY
Ann. Return (%)	9.53	8.87	10.28	10.42
Ann. Vol (%)	19.29	18.82	18.95	19.86
Sharpe Ratio	0.390	0.365	0.437	0.424
Max Drawdown (%)	-52.65	-47.32	-49.14	-55.19
Calmar Ratio	0.181	0.187	0.209	0.189

Table 1: Out-of-sample performance, 2007-2024 (net of 10 bps TC). Risk-free rate: 2% p.a.

Table 1 tells a consistent story across five dimensions.

Annualised return. (Total portfolio gain per year, compounded.) RS delivers the highest return among our three strategies (10.28%), just below SPY (10.42%). MS underperforms at 8.87%, which we attribute to the noise in sample mean return estimates: the max-Sharpe optimiser concentrates weights in sectors that happened to do well historically, but this does not persist out of sample.

Volatility. (Annualised standard deviation of daily returns.) All strategies are tightly clustered between 18.8% and 19.3%, just below SPY (19.86%). The 30% weight cap limits how much MV can reduce volatility by concentrating in defensive sectors, which explains the small spread.

Sharpe ratio. (Excess return per unit of volatility, risk-free rate at 2%.) RS achieves the best

Sharpe (0.437), ahead of SPY (0.424), MV (0.390), and MS (0.365). The result for MS is notable: despite having the lowest volatility of the three strategies, its poor return drags the Sharpe below even the passive benchmark.

Maximum drawdown. (Worst peak-to-trough loss observed over the full period.) MS has the shallowest drawdown (-47.32%), which at first seems inconsistent with it being the most aggressive strategy. The explanation is that MS tends to rotate away from sectors with deteriorating momentum, whereas MV holds its defensive weights rigidly and still suffers when all sectors fall together, as in 2008.

Calmar ratio. (Annualised return divided by maximum drawdown.) RS ranks first (0.209), which is the metric that best summarises its profile: decent return combined with a drawdown shallower than MV and SPY. MV ranks last (0.181) despite its reputation as a defensive strategy, because its return is too low to compensate for a drawdown almost as deep as SPY's.

3.3. Performance breakdown by regime

Regime	Days	Sharpe (MV)	Sharpe (MS)	Sharpe (RS)	Sharpe (SPY)
Bear	1011	-0.019	0.086	-0.036	-0.034
Neutral	2302	1.218	0.870	1.204	1.202
Bull	1133	-0.151	0.004	0.007	-0.094

Table 2: Sharpe ratios by detected market regime.

Table 2 is the key diagnostic table: it answers whether the regime-switching rule delivers what it promises in each sub-period.

Bear regime (1011 days). The most counterintuitive result of the report: MS (0.086) outperforms both MV (-0.019) and RS (-0.036) in bear quarters. One explanation is that the KMeans bear label does not correspond exclusively to sustained downturns. Some of these quarters are volatile but end flat or positive, and in those cases the higher-risk MS allocation does better. Overall, all strategies stay close to SPY (-0.034), which suggests that sector diversification provides limited protection when the whole equity market sells off.

Neutral regime (2302 days, 52% of the sample). This is the dominant regime and where the overall performance gap is built. MV (1.218) and RS (1.204) both clearly beat MS (0.870) and SPY (1.202). RS correctly holds MV weights in neutral quarters, and the near-identical Sharpe between the two (1.204 vs 1.218) shows that the switching decision adds almost no cost in normal market conditions.

Bull regime (1133 days). All strategies post weak or negative Sharpe ratios here, which requires explanation. The bull label is assigned by KMeans based on short-term momentum features, and some of these quarters turn out to be transitional periods before a correction. RS (0.007) and MS (0.004) are marginally positive since RS holds MS in bull quarters, but the difference is not meaningful. MV (-0.151) is the clear laggard, penalised by its underweight in high-momentum sectors during periods the model identifies as bullish.

3.4. Look-ahead bias

Strategy	Sharpe (in-sample)	Sharpe (walk-forward)	Gap
Min-Variance	0.443	0.390	0.053
Max-Sharpe	0.599	0.365	0.234

Table 3: Look-ahead bias: in-sample vs walk-forward Sharpe.

Table 3 shows how much performance is overstated when the covariance matrix and expected returns are estimated on the full sample and then evaluated on the same period. For MV the gap is small (0.443 vs 0.390, a difference of 0.053), because MV does not use mean return estimates and is therefore less sensitive to this source of error. For MS the gap is severe: the in-sample Sharpe of 0.599 collapses to 0.365 out of sample, a drop of 0.234 or nearly 40%. This confirms that the max-Sharpe strategy looks attractive in-sample precisely because it overfits to historical mean returns, which do not repeat. This is the main reason we use a strict walk-forward framework throughout.

4. Discussion and Limitations

The results confirm that regime-switching adds value on aggregate: RS achieves the best Sharpe (0.437) and Calmar (0.209) out of sample. However, the regime breakdown reveals that this outperformance is concentrated in neutral periods, not in bear markets as originally intended. The classifier’s lag and the mixed composition of “bear” and “bull” clusters limit the strategy’s ability to time regime transitions precisely.

Our key limitations are the following: (1) KMeans does not model regime persistence: a Hidden Markov Model would better capture the stickiness of bear markets; (2) sample covariance without shrinkage amplifies estimation error, particularly for MS; (3) the universe is limited to U.S. equity sectors, where cross-sector correlations spike during crises, reducing diversification benefits.

(4) Moreover, regime detection relies solely on market microstructure features (realised volatility, momentum, VIX), capturing only the equity market’s own signal. A more robust framework would incorporate macroeconomic state variables such as the yield curve spread, inflation, and commodity prices. Mulliner, Harvey et al. [4] formalise this approach by measuring how closely current conditions resemble historical epochs rather than imposing predefined labels, and show that “anti-regimes” (periods maximally dissimilar to today) also carry predictive value. Such a macro layer would allow the classifier to distinguish, for instance, a rate-shock episode from a credit crisis - two environments that look similar in VIX space but call for different allocations.

5. Conclusion

This report presents a walk-forward sector allocation framework combining Markowitz optimisation with unsupervised regime detection. The regime-switching portfolio achieves the best risk-adjusted performance overall (Sharpe 0.437, Calmar 0.209), driven primarily by its strong neutral-regime performance (Sharpe 1.204) and its superior overall return-to-drawdown profile. The look-ahead bias analysis further highlights that Max-Sharpe in-sample Sharpe overstates the

walk-forward result by 0.234 units-a cautionary result for any practitioner relying on in-sample optimisation. Natural extensions include Ledoit-Wolf covariance shrinkage, Hidden Markov Model regime detection, and a macro-regime layer based on the 10Y-2Y yield curve spread [3].

References

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